1 A computer has a Central Processing Unit (CPU).

Describe what happens during the fetch-execute cycle.

[2]

2 A computer's CPU contains registers.

Complete the table by identifying **two** registers **and** state the purpose of each register.

Register	Purpose

[2]

3(a) A computer has a Central Processing Unit (CPU).

Complete the table by writing the name of **two** registers used in the fetch-execute cycle **and** the purpose of each register.

Register	Purpose

[4]

(b) Give **three** characteristics of a CPU that can affect its performance.

1

2

3 [3]

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

4 A car has a ‘Follow Me’ system that uses a cruise control feature to allow the car to follow the car in front of it. It will keep the same speed and distance without the driver’s intervention. The cruise control system is an example of an embedded system.

Explain the reasons why the ‘Follow Me’ system is an example of an embedded system.

[3]

5 A car comes with many embedded systems, for example parking sensors.

Identify **one** other embedded system that could be found in a car and explain why this is an embedded system.

Example embedded system

Explanation

[3]

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

6 Complete the table by writing the missing definition or name of each of the common CPU components and registers.

CPU component or register	Definition
	Stores the address of the next instruction to be fetched from memory. Increments during each fetch-execute cycle.
CU (Control Unit)	
	Stores the address of the data to be fetched from or the address where the data is to be stored.
	Performs mathematical calculations and logical operations.

[4]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			1 mark each • **Data/instructions** are fetched from **memory/RAM/primary storage** • **Data/instructions** are stored using the registers // **correct example** of a register storing address/data • **Data/instructions** are decoded // **Data/instructions** are split into opcode and operand • **Data/instructions** are **executed/processed** • **ALU** performs the logical/arithmetic calculations	2	MP4 BOD carried out etc. for executed. Ignore inaccurate references to registers and components (other than MP2 correct example of a register). <u>Examiner's Comments</u> Candidates often correctly identified that data is fetched from memory, or from RAM, and are then processed. Some candidates gave a more technical description including the role of the registers in this process. The stronger responses included clear references to data or instructions being processed. Some candidates inaccurately identified that information was processed, or that programs were fetched from memory.
			Total	2	
2			1 mark for register with purpose • Program counter // PC **Stores** the **address** of the **current/next** instruction to be fetched // **stores** the **address** of the instruction for the **current/next** FE **cycle** • Memory address register // MAR **Stores** the **address** of the **current/next/required instruction/data** // **stores** the **address** of **data/instruction** about **to be fetched/executed** // **stores** the **address** where **data/instruction** (in MAR) **is going to be stored** // **stores** the **address** of **instruction/data being decoded/executed** • Memory data register // MDR **Stores** the **data/instruction** fetched from memory // **stores data/instruction** to be stored in memory // stores the **data/instruction** located in the **memory** location in the **MAR** • Accumulator // ACC **Stores** the **result** of calculations // **stores data currently** being processed / by example // **stores** the result from the ALU	2	Careful that the purpose is not an action such as fetches, takes, retrieves. Accept "points" to/at in place of stores Accept • Current instruction register//CIR//Instruction register//IR Stores the instruction currently being executed BOD memory buffer register for MDR. Accept memory address, memory data without 'register'. <u>Examiner's Comments</u> Many candidates were able to identify registers that are in the CPU, most commonly the MAR and MDR. Some candidates inaccurately identified other components of the CPU such as the ALU and CU. Candidates were often able to gain the mark for a description of the MDR. Some candidates inaccurately identified the MAR or the PC as storing data instead of the address of the data. Candidates who gave the accumulator sometimes gave a response that it stored the calculations instead of the result of the calculations.

© OCR 2026. You may photocopy this page.

Created in ExamBuilder

Question			Answer/Indicative content	Marks	Guidance
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
3	a		1 mark for naming register, 1 for matching purpose • Program counter // PC • **Stores** the **address** of the **current/next** instruction to be fetched // **stores** the **address** of the instruction for the **current/next** FE **cycle** • Memory address register // MAR • **Stores** the **address** of the **current/next** instruction/data to be fetched // **stores** the **address** where data/instruction is to be stored • Memory data register // MDR • **Stores** the **data/instruction** fetched from memory // **stores data/instruction** to be stored in memory // stores the **data/instruction** located in the **memory** location in the **MAR** • Accumulator // ACC • **Stores** the **result** of calculations // **stores data currently** being processed / by example // **stores** the result from the ALU	4	Careful that the purpose is not an action such as fetches, takes, retrieves. Read full purpose and award a correct point Accept • Current instruction register//CIR//Instruction register//IR • Stores the instruction currently being executed BOD memory buffer register for MDR. If there is no register but the register is given in the purpose column, award the purpose if accurate. If the answer in the register column is incorrect, do not mark purpose. For PC and MAR, accept 'pointer' for storing address Accept memory address, memory data <u>Examiner's Comments</u> Candidates were often able to identify one or two registers that are used in the F-E cycle. Fewer candidates were able to give a purpose in the F-E cycle. Some candidates identified that the registers were involved in the fetching or transmission of data, for example that the MAR transmits the address to RAM.. **Misconception** A common misconception is that the program counter keeps track of how many programs have run or counts the instructions that are being processed.

Question			Answer/Indicative content	Marks	Guidance
	b		1 mark each to max 3 • Clock speed • Cache size • Number of cores	3	'clock' 'cache' 'speed' 'cores' on its own is NE. <u>Examiner's Comments</u> Candidates were often able to identify at least one characteristic of a CPU, most commonly the clock speed and number of cores. Some responses were not precise enough as to the characteristics, for example stating 'clock' or 'core' without reference to the speed of the clock, or the number of cores, which were too ambiguous.
			Total	7	
4			1 mark each to max 3 • Has a specific purpose // it only performs one/limited task // dedicated to the Follow Me system • Built within a larger device/car • **Dedicated**/specific/its own hardware / sensors • Has a microprocessor • Built-in operating system/software // software is all in firmware/ROM • … it's instructions/operation does not/is hard to change/update • It is a control system // it is automated	3	MP2 BOD reference to it being 'built into' 'something' reasonable <u>Examiner's Comments</u> This question required candidates to apply their understanding of embedded systems to a different system. Candidates were often able to identify the key features of embedded systems that were relevant to this scenario. The most common points being that the system has a single purpose. Some candidates also identified that the system is built within a larger system, being the car. Fewer candidates were able to provide a third point. Those that did most commonly identified the dedicated hardware or gave an example such as the sensors are only providing data for this system.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
5			1 mark for example: e.g. • Auto lights • Auto window wipers • Sat nav // GPS • Airconditioning // climate control • Radio/entertainment/infotainment system/media system • Lane assist • Engine management system • Auto-park • Cruise control • Auto-brake • Follow-me • Dashcam 1 mark each to max 2 for explanation. • **Limited** functions // by example e.g. the system **only** checks the light and turns lights on/off • **Dedicated** microprocessor // by example e.g. there is a microprocessor that is only checking the lights • Hard to change function // by example e.g. the user cannot make the light system do any other role	3	Allow anything that could be reasonably within a car. If example is not clear if it's an embedded system, read explanation for justification e.g. hazard lights – could be embedded if they are activated automatically when the car crashes. Award the example in the explanation if this occurs. If justification is generic features of an embedded system max 1 for explanation. Do not award 'built into the car/larger machine' because this is in the question. <u>Examiner's Comments</u> This question required candidates to consider embedded systems within a car. There were a range of possible systems, the most common being GPS or satellite navigation systems. Other common responses included automated lights, automated wipers, and parking sensors. The most common explanation was that the system has a single (or limited) purpose, but few candidates expanded beyond this. Some candidates repeated that it was built into the car but this was provided in the question. Some candidates provided examples of embedded systems such as a washing machine, a microwave and a fridge/freezer. This was not appropriate to the context of the question.
			Total	3	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>6</td><td></td><td></td><td>1 mark for each term or definition
<table>
<tr><th>CPU component or register</th><th>Definition</th></tr>
<tr><td>Program Counter // PC</td><td>Stores the address of the next instruction to be fetched from memory. Increments in each fetch-execute cycle.</td></tr>
<tr><td>CU (Control Unit)</td><td>(Sends signals to) synchronise / control / coordinates the processor/hardware/F-E cycle/processes/flow of data // decodes instructions (in CIR)
// runs F-E cycle</td></tr>
<tr><td>Memory Address Register // MAR</td><td>Stores the address of the data to be fetched from, or the address where the data is to be stored.</td></tr>
<tr><td>Arithmetic Logic Unit // ALU</td><td>Performs the mathematical and logical calculations.</td></tr>
</table></td><td>4</td><td>Read whole answer for CU and award correct point at any stage.

CU ‘sends signals to components’ is not enough, it isn’t saying what the signal’s purpose is

<u>Examiner’s Comments</u>

This question required candidates to consider the definitions and identify the component or register that was being defined, as well as giving a definition for the CU.

The ALU was often correctly identified.

Common errors included giving MAR for the first definition in place of the PC and then following on with the MDR for the third row because the MAR had already been given by the candidate.

Many candidates demonstrated a good understanding of the CU, most commonly identifying that it controls the flow of data, that it controls the FDE cycle or that it decodes instructions.

Misconception

A common error was stating that the Control Unit actually performs the FDE cycle, or that it executes the instructions.</td></tr>
<tr><td></td><td></td><td></td><td>Total</td><td>4</td><td></td></tr>
</table>